



ECON 100B: Macroeconomics

Class 17: The IS Curve [2/2] (Jones Chapter 11)

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March 31, 2026

Agenda

- Announcements
- Current events
- Finish Chapter 11: The IS Curve

Announcements

- **Problem Set 3** will be available soon, due on **Friday, April 17**
- **Problem Set 4** will be due on **Friday, May 1**
- The **Final exam** is confirmed for **Monday, May 11**
 - Just like at midterm, the main class meets here in Pimentel 1
 - We will run 90 minutes from about 3:00-4:30PM
 - Like at midterm, DSP students please either make a proctoring appointment, or for 150% time only, we will get another classroom

Current events

- On [April Fools' Day](#) but no joke: On Wednesday, 4/1/26, the U.S. Supreme Court hears oral arguments in [Trump v. Barbara](#)
 - At issue is an [executive order](#) from January 2025 and the [1898 ruling in U.S. v. Wong Kim Ark](#) holding that the 14th Amendment granted citizenship upon being born in the U.S.
 - Wong Kim Ark was born in San Francisco's Chinatown in the 1870s to unauthorized immigrant parents from China
- [Chung, Andrew \(3/29/26\) "In Supreme Court fight over birthright citizenship, a great-grandson hears echoes of 1898," Reuters](#)
- "Barbara" is the first name reportedly of a Honduran national. The case is a nationwide class-action suit brought by the ACLU & others
- SCOTUS is expected to rule this summer

More current events

- [Nick Timiraos \(3/30/26\) "Powell Says Fed Can Look Past Oil Shock, but Warns Patience Has Limits," WSJ](#)

- On Monday, March 30, Federal Reserve Chair Jerome Powell addressed students at Harvard

- Lays out the dilemma:

An oil shock raises prices but might reduce growth; which one does the Fed try to fix?

- Answer: For now, wait and see
 - Energy disruptions tend to be short-lived
 - Monetary policy works too slowly to counteract them in real time
 - But five years of above-target inflation makes it harder to assume the public will simply shrug off another round of rising prices

Where we are and where we're headed

Chapter 9: The Short Run

Short-run output $\tilde{Y} = (Y - \bar{Y})/\bar{Y}$

Chapter 10: Historical Macro-clysms

$MV = PY$ and when M collapses, $\pi < 0 \rightarrow R > i$

Chapter 11: The IS Curve

$\tilde{Y} = \bar{a} - \bar{b}(R - \bar{r})$ because I falls when R rises

Chapter 12: Monetary Policy

The central bank chooses R in the short run

and the Phillips Curve

$\Delta\pi = \bar{v} \tilde{Y}$ because inflation increases when $\tilde{Y} > 0$

Chapter 13: Stabilization Policy

Central banks have an inflation target like $\bar{\pi} = 0.02$

Chapter 14: Financial Crises

Like deflation, risk premia can also push R too high

Chapter 18: Government

Government can stabilize demand but can't spend forever

Chapter 19: International Trade

Trade deficits are not all they seem; $S - I = NX$

Chapter 20: Exchange Rates

Rising R attracts foreign money, causes appreciation, $NX \downarrow$

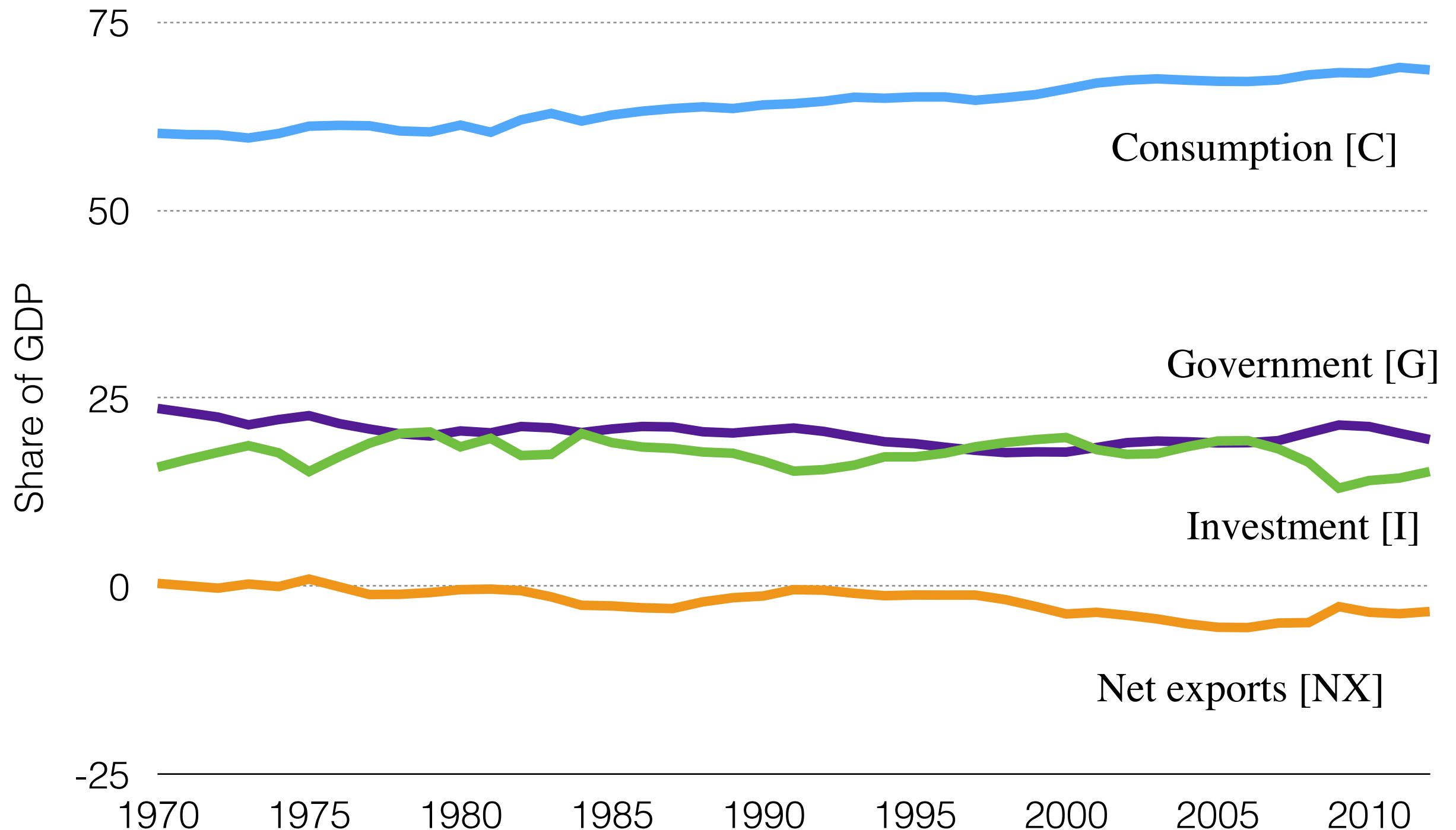
Learning objectives for Chapter 11

- Think of $Y = C + I + G + EX - IM$ as **aggregate demand**
- The IS curve shows how short-run output $\tilde{Y}$...
 - Is a downward-sloping function of the real interest rate R
 - Shifts inward when components of aggregate demand fall, and shifts outward when components of aggregate demand rise
- Several expenditure components might be sensitive to the interest rate, but we focus on **investment**
 - A higher real interest rate R is an increased price of spending today
 - (ECON 100A:) For many economic agents, the substitution effect of a higher R outweighs any income effect, and spending today declines when R rises
- The math of the IS curve is a downward sloping line: $\tilde{Y} = \bar{a} - \bar{b} (R - \bar{r})$

Chapter 11 part 2 (today)

- Derivation again, quickly
- Calibration — real values — and numeric examples
- Why it's called the IS curve, which takes us into math identities
- A first look at a very unintuitive but important identity: $S - I = NX$
- Remember the Keynesian Cross from ECON 2?
 - If consumers
 - Spend a share of extra income ...
 - Reduce spending when income falls ...
 - Then the IS curve becomes flatter. Shocks and fiscal and monetary policies are all **amplified**

The expenditure shares of GDP have been stable,
except for investment



The IS curve shows how investment spending I and short-run output $\tilde{Y}$ are decreasing in the real interest rate R

$$Y = C + I + G + NX$$

Start with expenditure-side GDP

The IS curve shows how investment spending I and short-run output $\tilde{Y}$ are decreasing in the real interest rate R

$$Y = C + I + G + NX$$

$$\frac{Y}{\bar{Y}} = \frac{C}{\bar{Y}} + \frac{I}{\bar{Y}} + \frac{G}{\bar{Y}} + \frac{NX}{\bar{Y}}$$

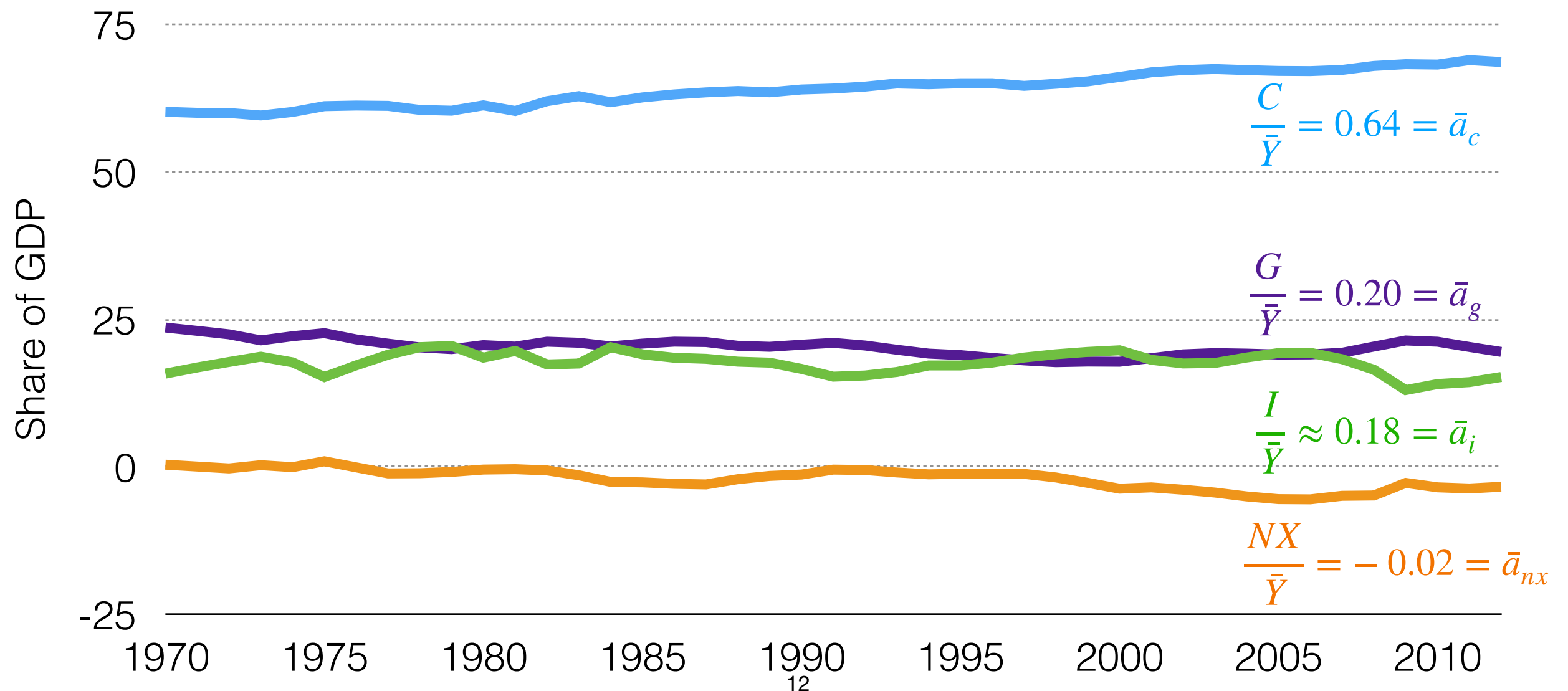
Divide both sides by $\bar{Y}$

The IS curve shows how investment spending I and short-run output $\tilde{Y}$ are decreasing in the real interest rate R

$$Y = C + I + G + NX$$

$$\frac{Y}{\bar{Y}} = \frac{C}{\bar{Y}} + \frac{I}{\bar{Y}} + \frac{G}{\bar{Y}} + \frac{NX}{\bar{Y}}$$

Economic agents set most of these equal to constant shares



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$$\frac{Y}{\bar{Y}} = \frac{C}{\bar{Y}} + \frac{I}{\bar{Y}} + \frac{G}{\bar{Y}} + \frac{NX}{\bar{Y}}$$

$$\frac{Y}{\bar{Y}} = \frac{Y - \bar{Y}}{\bar{Y}} + \frac{\bar{Y}}{\bar{Y}} = \tilde{Y} + 1$$

The math gets a little weird here

Notice how the left-hand side equals $\tilde{Y} + 1$

The IS curve shows how investment spending I and short-run output $\tilde{Y}$ are decreasing in the real interest rate R

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$$\frac{I}{\bar{Y}} = \bar{a}_i - \bar{b}(R - \bar{r})$$

$$\tilde{Y} = \bar{a}_c + [\bar{a}_i - \bar{b}(R - \bar{r})] + \bar{a}_g + \bar{a}_{nx} - 1$$

If investment falls with R , we arrive here

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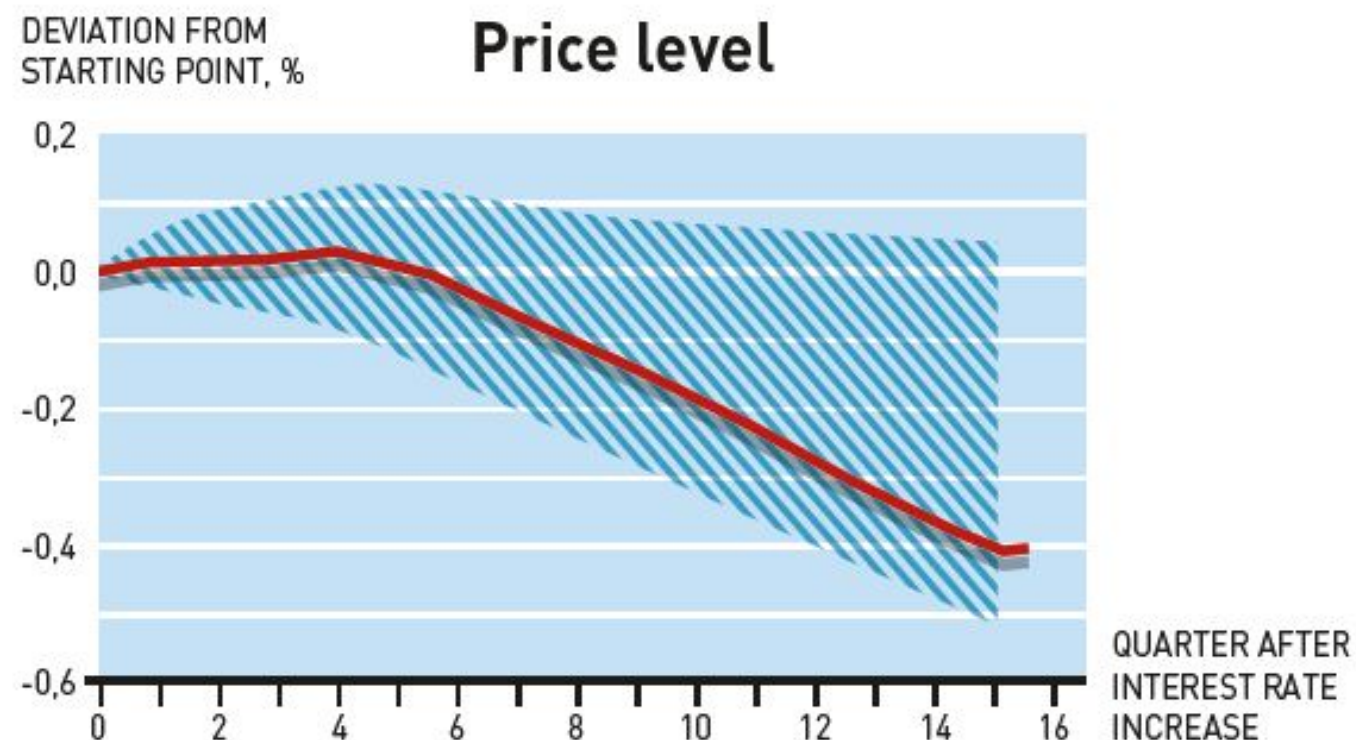
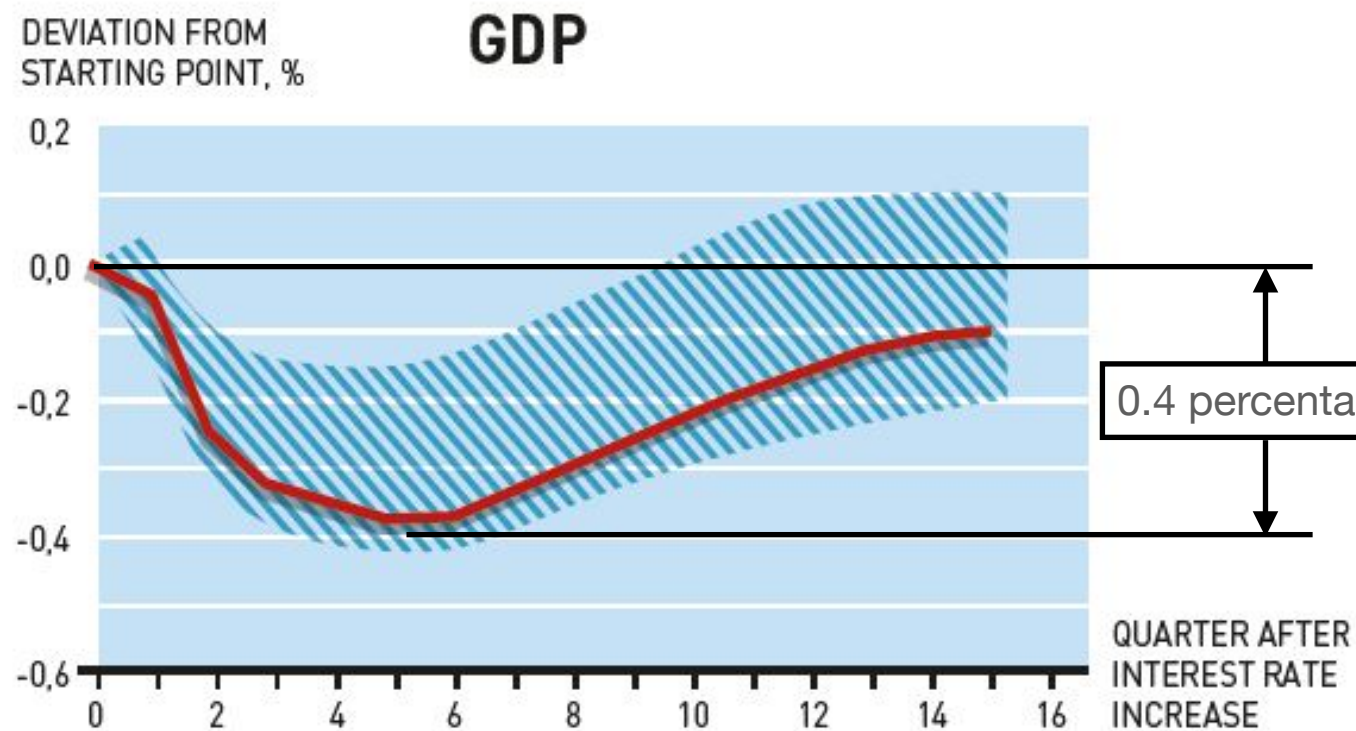
All the $\bar{a}_j$'s work the same way. Let $\bar{a} \equiv \sum \bar{a}_j - 1$

$$\tilde{Y} = \bar{a} - \bar{b} (R - \bar{r})$$

This is the IS Curve. Simple.

**Let's calibrate the model with
Sims's VAR from Class 16**

From the Nobel page, a monetary policy VAR



- What do the data say about the effects of a 1 percentage point increase in the target federal funds interest rate?
- After 4-6 quarters or about a year, when R rises by 1 percentage point, GDP falls by 0.4 percentage point

Let's calibrate the IS curve with Sims's VAR from Class 16

- In Chris Sims's VAR, if $R - \bar{r} = 0.01$ then after 4 quarters,

$$\tilde{Y} = -0.004$$

- In words:

When the Fed raises the real interest rate by 1 percentage point above its long-run rate $\bar{r}$,

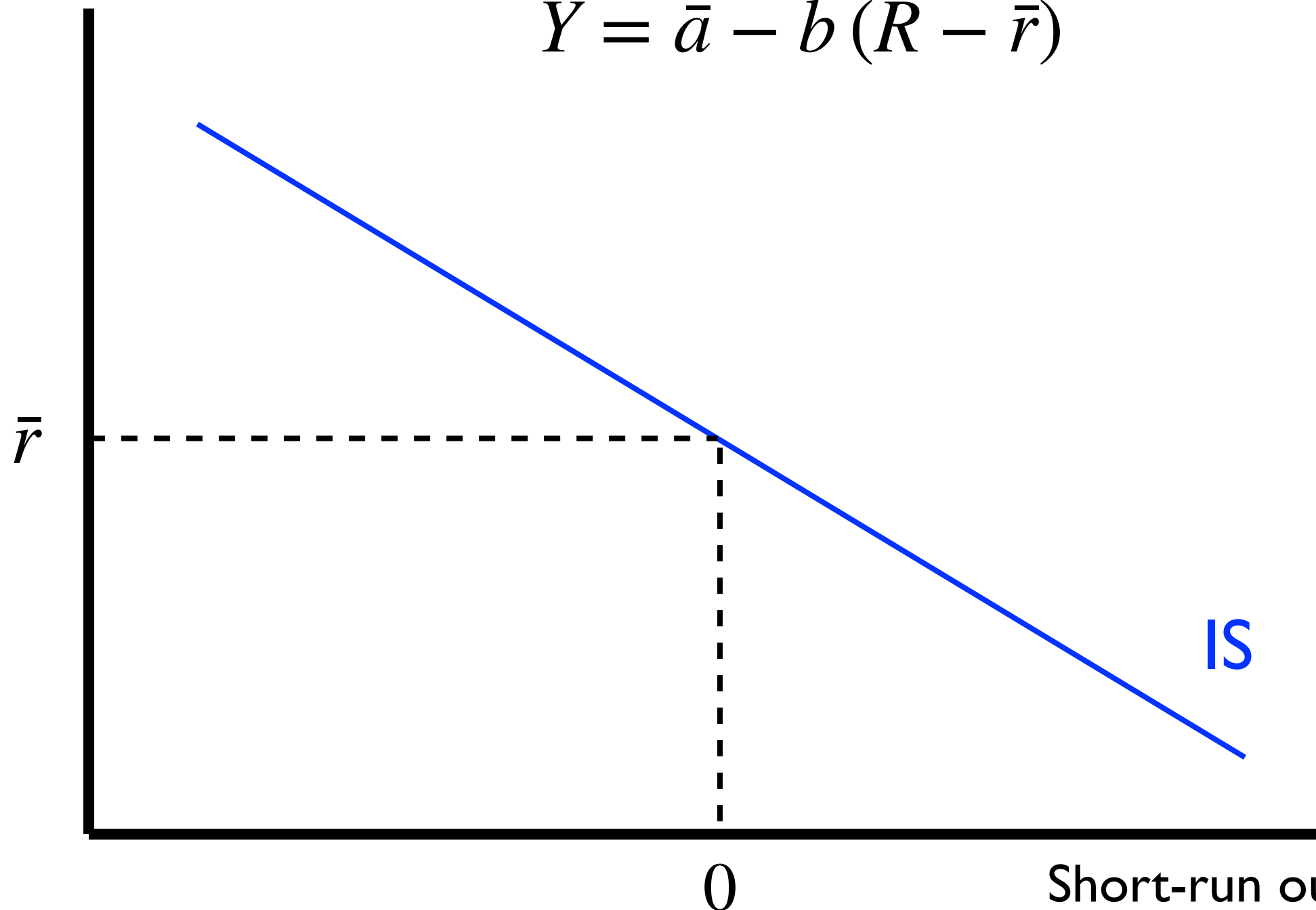
then output falls 0.4 percentage point below potential

- This fits

$$\bar{b} = 0.4 = \frac{2}{5}$$

Real interest rate, R

$$\tilde{Y} = \bar{a} - \bar{b} (R - \bar{r})$$

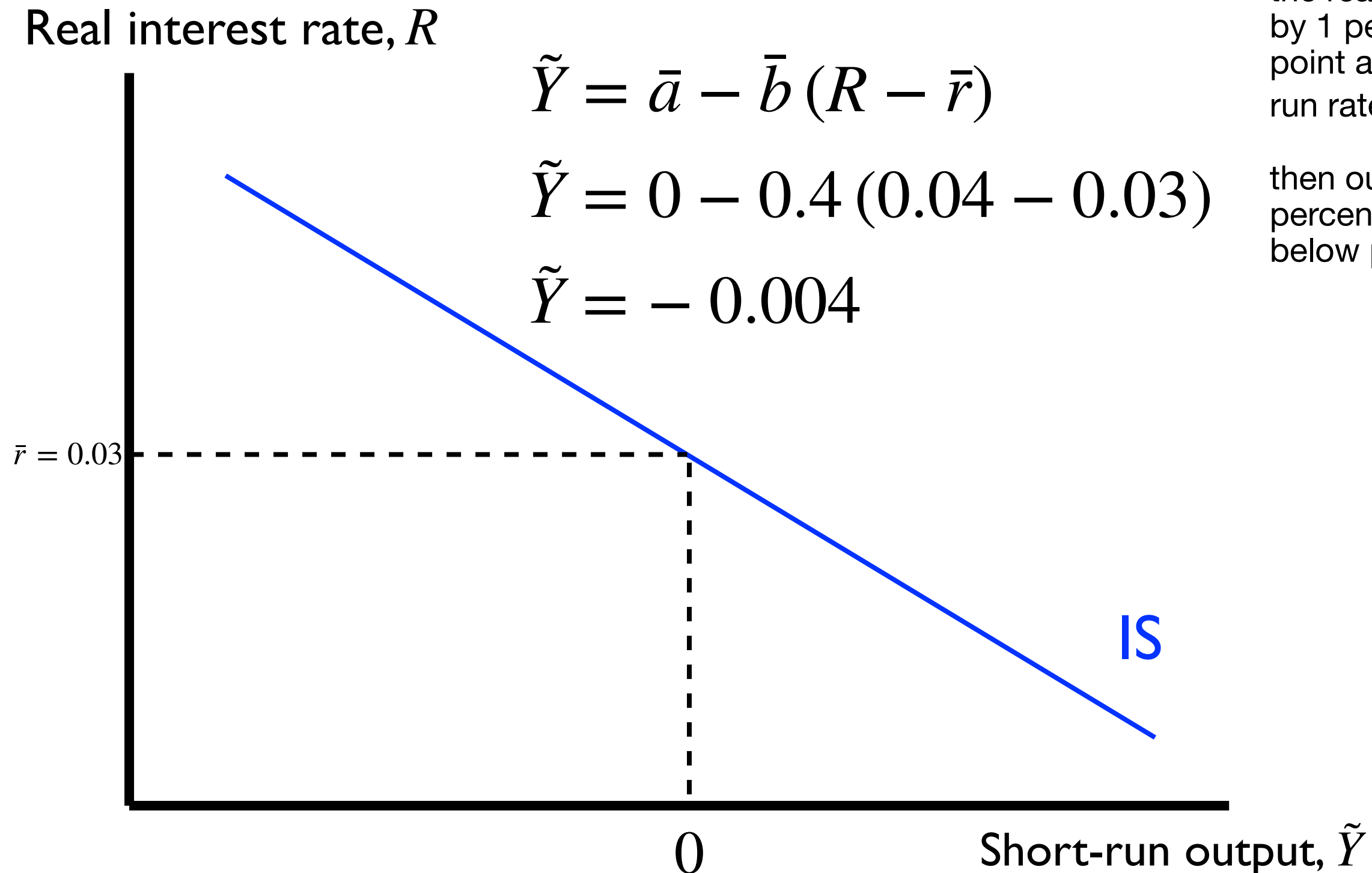


Without any demand shock, $\bar{a} = 0$

The risk-free marginal product of capital might be $\bar{r} = 0.03$

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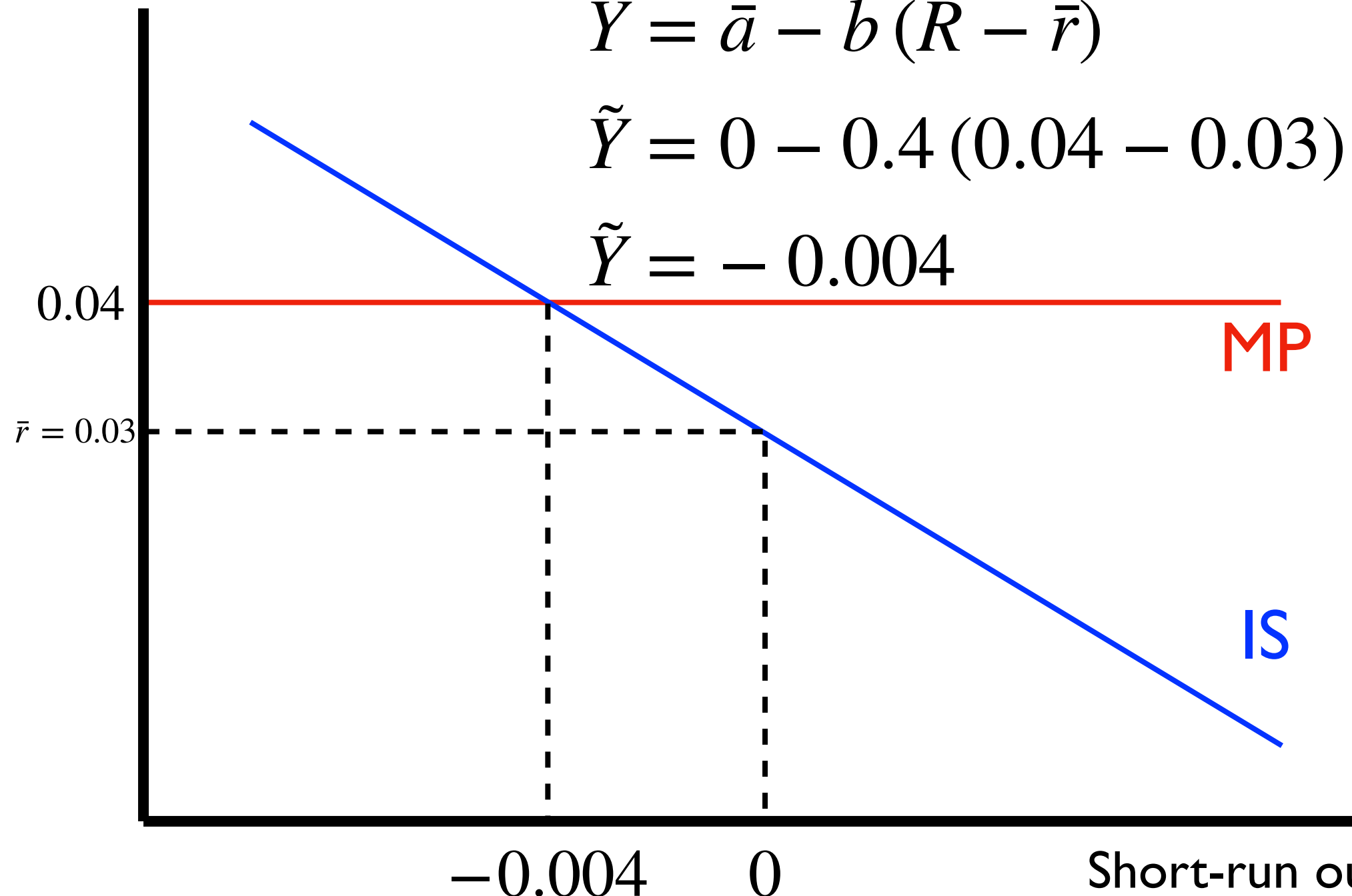
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Real interest rate, R

$$\tilde{Y} = \bar{a} - \bar{b} (R - \bar{r})$$

$$\tilde{Y} = 0 - 0.4 (0.04 - 0.03)$$

$$\tilde{Y} = -0.004$$



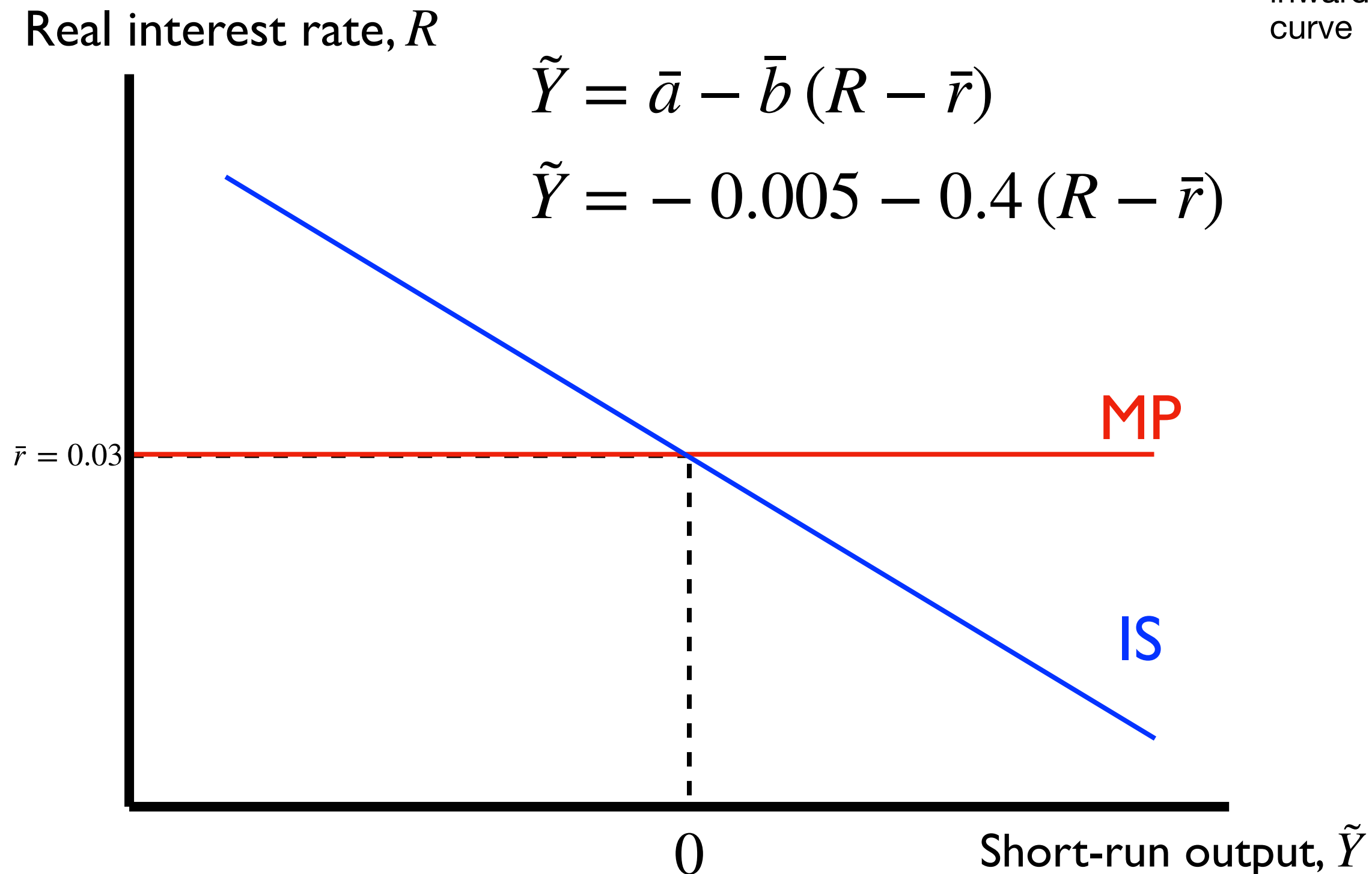
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- We will show the level of R that is chosen by the central bank with a flat **MP curve**

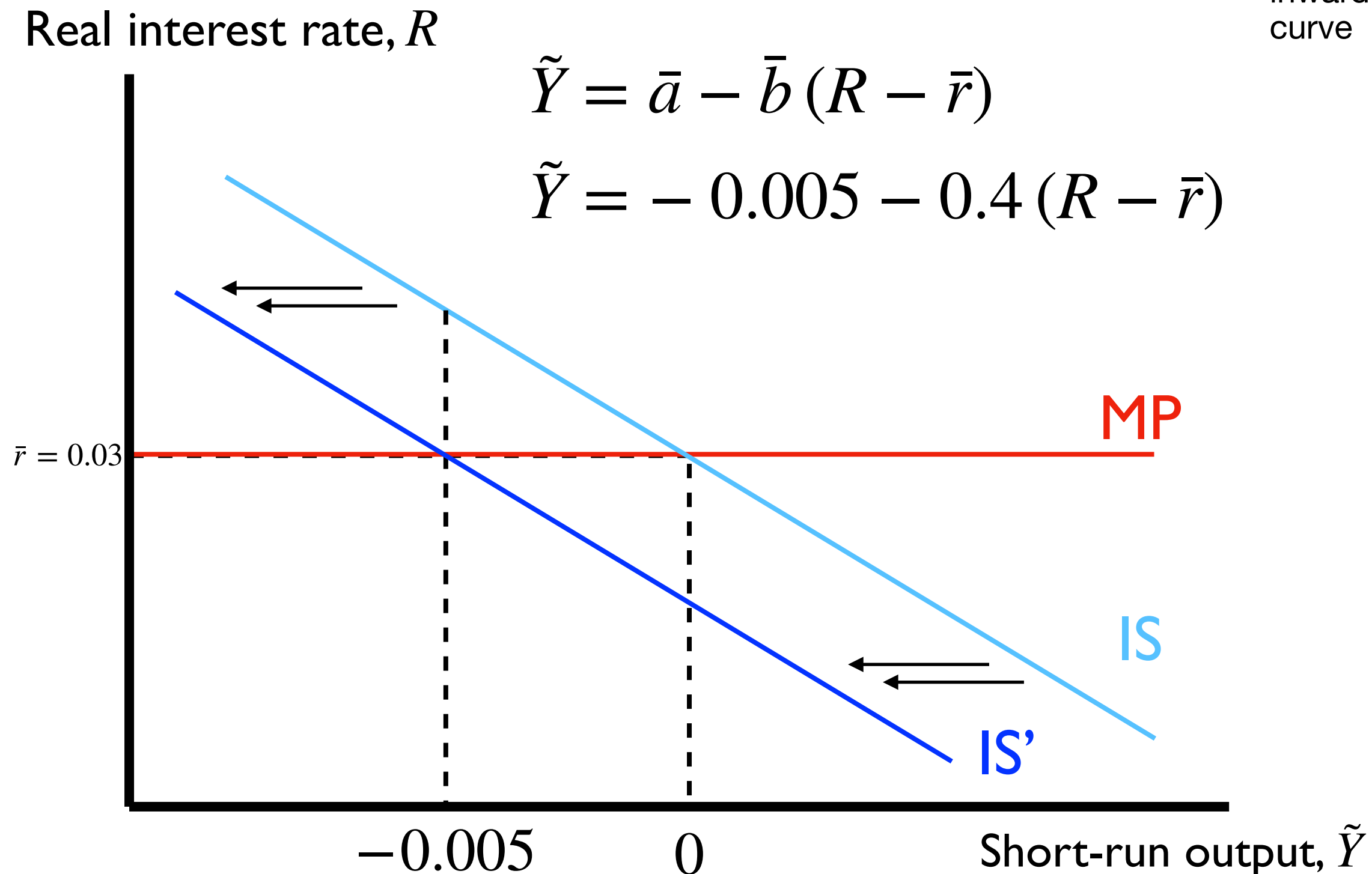
Now suppose monetary policy is neutral, with $R = \bar{r} = 0.03$ but there is a negative demand shock of 0.5% that temporarily reduces spending, decreasing from $\bar{a} = 0$ to $\bar{a}' = -0.005$

- This is a leftward or inward shift of the IS curve

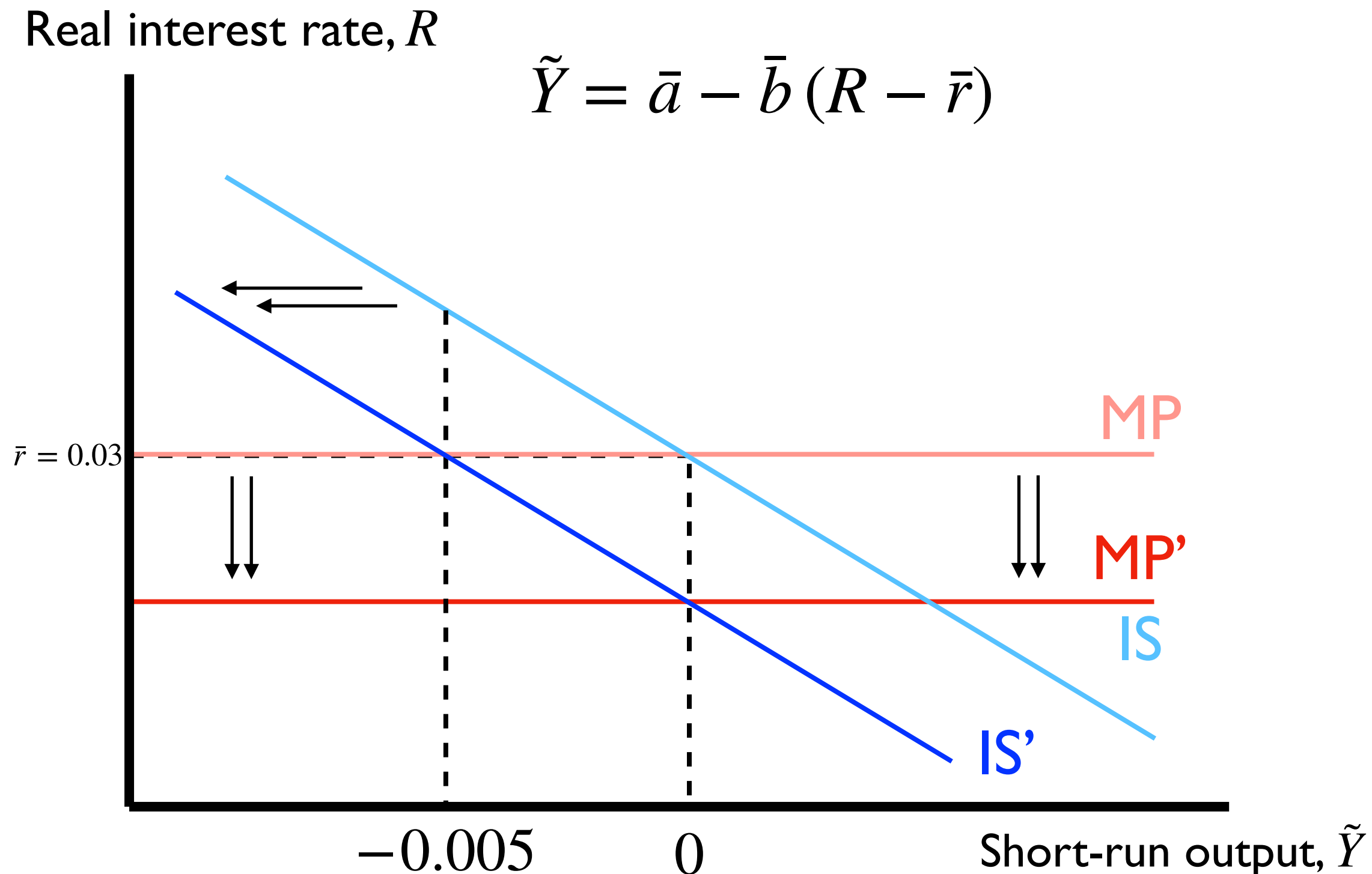


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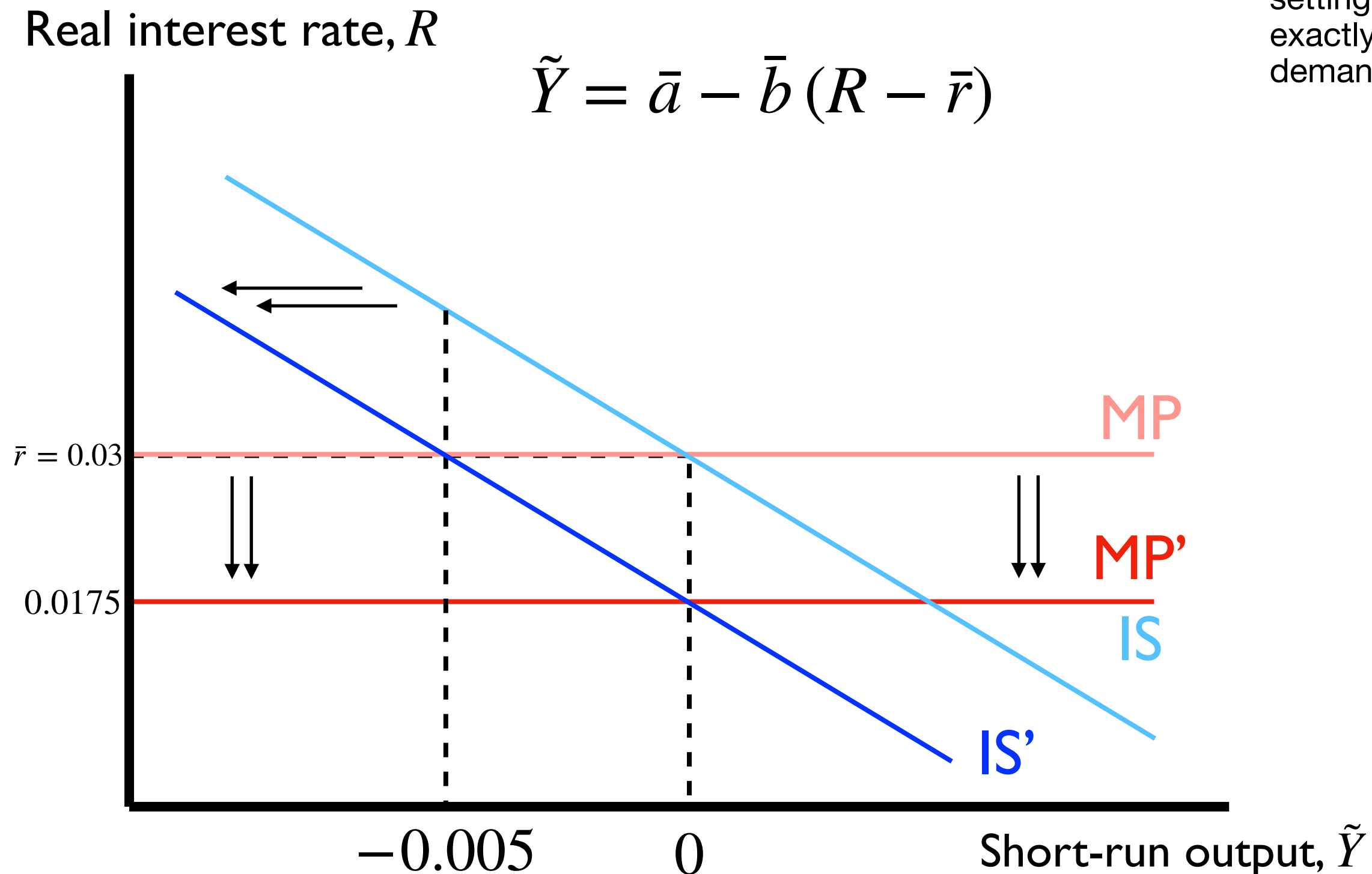


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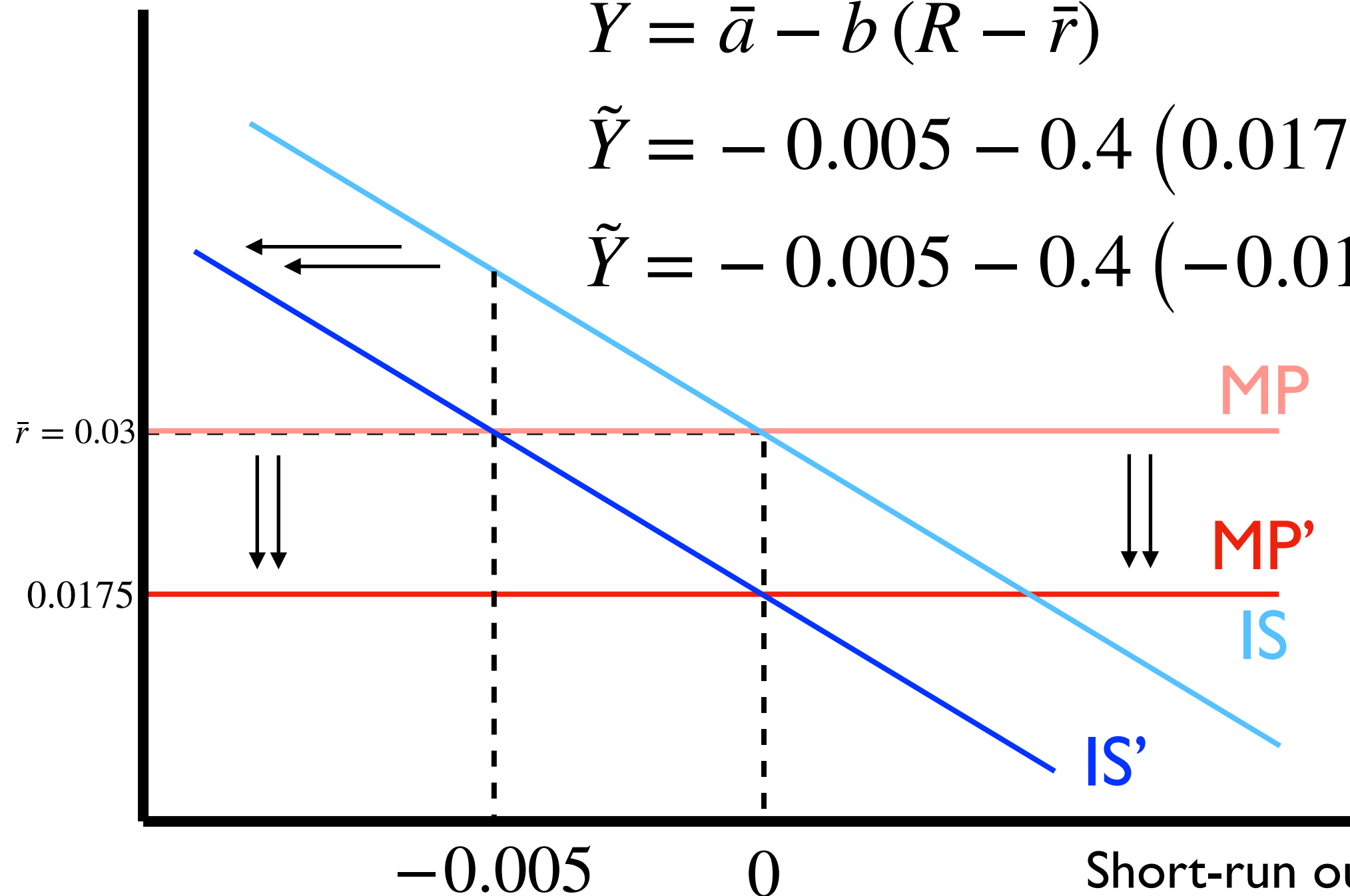
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Real interest rate, R

$$\tilde{Y} = \bar{a} - \bar{b} (R - \bar{r})$$

$$\tilde{Y} = -0.005 - 0.4 (0.0175 - 0.03)$$

$$\tilde{Y} = -0.005 - 0.4 (-0.0125) = 0$$



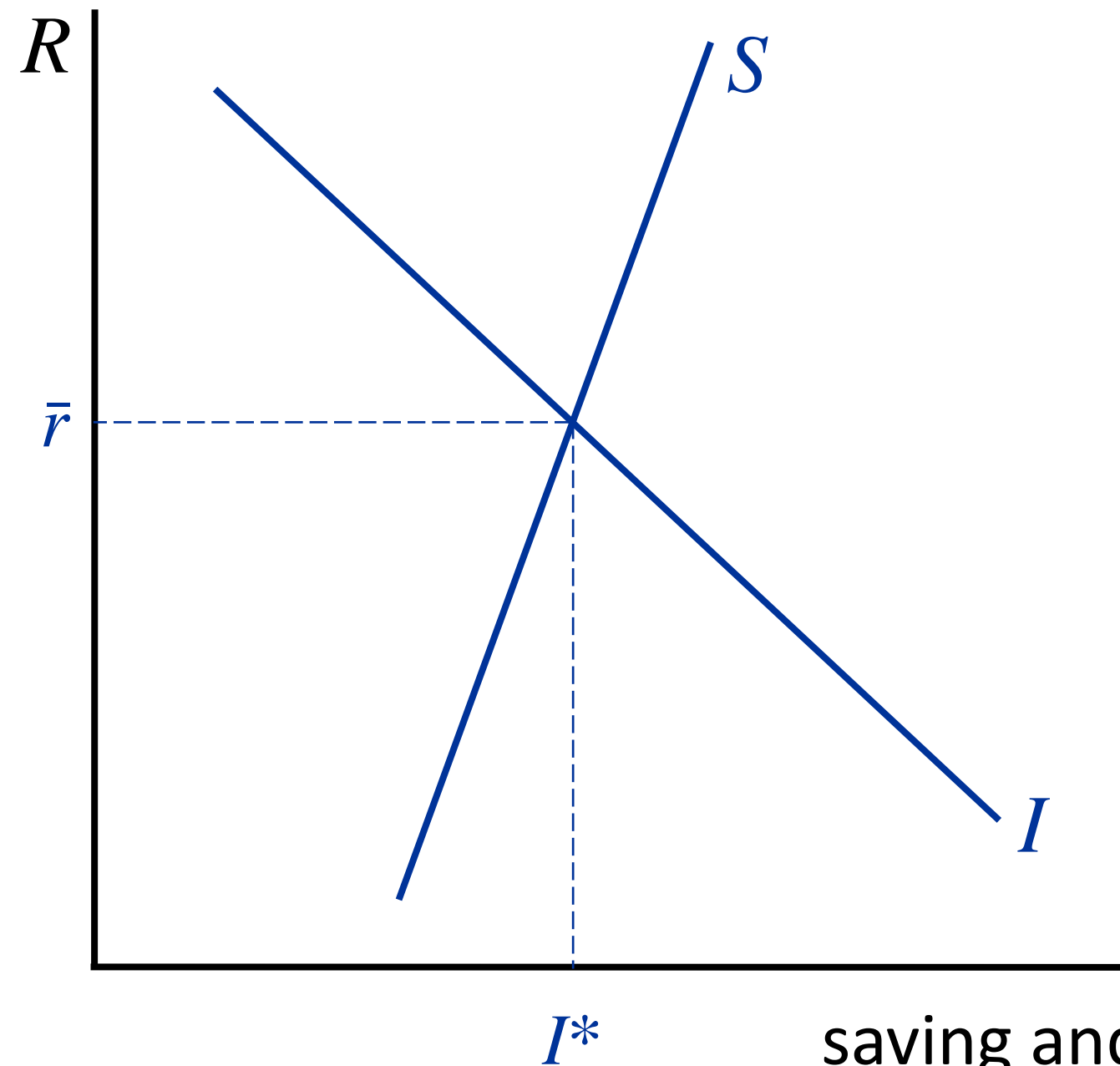
- (As we will see, the demand shock may have reduced the rate of inflation below the target, so the central bank might prefer $\tilde{Y} > 0$ temporarily)

Next topic: Why is it called the “IS curve” anyway?

- In simple markets with demand and supply, the price adjusts so that demand = supply
- In the macroeconomy, it's also useful to think of an equality that results when a price adjusts
- INVESTMENT equals SAVING when the interest rate R clears the market

Great. So how do we get started?

In ECON 2, Class 19 Slide 17 posits a long-run relationship between saving and investment



Investment demand slopes down like a demand curve with the price of investment equal to the real interest rate R

Saving is the supply of funds for investment, and there's probably some upward slope



In ECON 100B, use what you know

- Saving is what's left out of income after consumption and taxes

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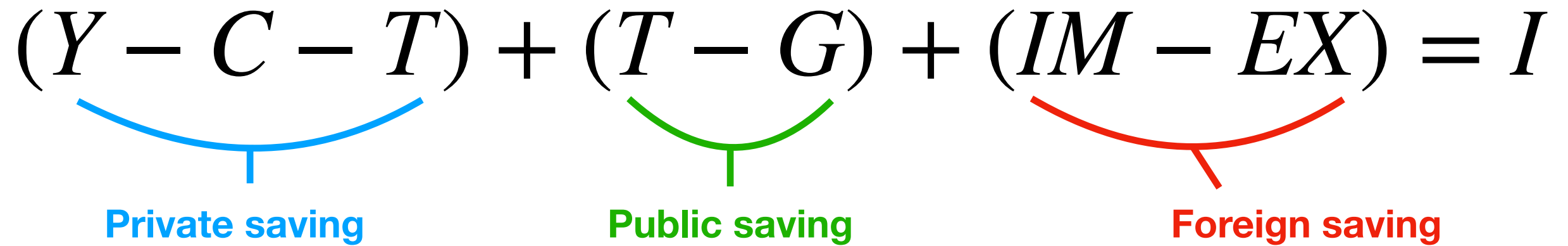
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Private saving

Public saving

Foreign saving

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Private saving

Public saving

Foreign saving

- Domestic investment spending must be financed with something.

$$(Y - C - T) + (T - G) + (IM - EX) = I$$

Private saving
Public saving
Foreign saving

- Domestic investment spending must be financed with something.
- Here are the three Somethings, for the two Somebodies: Us and Them
 - **Private saving** Private individuals can spend less C and save more
 - **Public saving** Government can spend less G and save more
 - **Foreign saving** This is weird. We can import more IM from Them or we can export less EX to Them

Hint: Think of imports IM as being primarily capital goods

Last topic:

Spending or saving extra income



Department of the Treasury
Internal Revenue Service

Notice 1275 (June 2001)
Catalog Number 31980B

www.irs.gov

Notice of Status and Amount of Immediate Tax Relief

Dear Taxpayer:

We are pleased to inform you that the United States Congress passed and President George W. Bush signed into law the Economic Growth and Tax Relief Reconciliation Act of 2001, which provides long-term tax relief for all Americans who pay income taxes.

The new tax law provides immediate tax relief in 2001 and long-term tax relief for the years to come.

As part of the immediate tax relief, you will be receiving a check in the amount of

\$300.00 during the week of 08/13/2001.



Department of the Treasury
Internal Revenue Service
Andover, MA 05501-0025

Notice Date: May 12, 2008

Notice Number: CP 1378

Taxpayer Identification Number:

Primary: XXX-XX-1234

John Q. & Jane R. Public
100 Main Street
Springfield, CA
12345

For assistance, you may call:

1-866-234-2942

Understanding Your Economic Stimulus Payment

Please keep a copy of this notice for your records.

Dear Taxpayer:

Your Economic Stimulus Payment

You are entitled to an economic stimulus payment of \$235.65 as provided by the Economic Stimulus Act of 2008. You can expect your payment by 5/9/08. If you do not receive it within six weeks of this notice, please contact us at the number shown above. You will not be required to report the amount of your stimulus payment as taxable income on your 2008 federal income tax return. If you receive any federal benefits or federally financed benefits, those benefits generally will not be affected by any stimulus payment you receive.

This really happened in 2001 and again in 2008 and in 2020

- In these years, the federal government sent tax rebates to U.S. households
- In 2001, singles in households got \$300 (like me), while couples in households got \$600
- In 2008, singles got between \$300 and \$600
couples got between \$600 and \$1200

But those amounts depended on your income; households with more income were given less

Same in 2020

How many people like to “blow it all?”

- It depends a lot on whom you’re talking about: everyone in the economy, or specific groups — also *when*, for example, during a crisis, or during normal times?
- Some people of relatively more limited means definitely live more “hand-to-mouth”
- For the U.S. economy as a whole, folks who “blow it all” might represent between 20% and 50%, let’s say 40%
- So approximately that share, $mpc = 0.4$, gets spent out of income fluctuations
- This mpc is the “marginal propensity to consume”

What are macroeconomic implications of “blowing it all?”

- When people spend a share $mpc = 0.4$ of their incomes, then income fluctuations **become multiplied**, or magnified

- Suppose instead of $\frac{C}{\bar{Y}} = \bar{a}_c$ we have $\frac{C}{\bar{Y}} = \bar{a}_c + mpc \cdot \tilde{Y}$

where mpc is a number between 0 and 1, like $mpc = 0.4$

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- Then we will find that the IS curve evolves from

$$\tilde{Y} = \bar{a} - \bar{b}(R - \bar{r}) \quad \text{to} \quad \tilde{Y} = \frac{1}{1 - mpc} [\bar{a} - \bar{b}(R - \bar{r})]$$

$$\text{where } \frac{1}{1 - mpc} = \frac{1}{1 - 0.4} = \frac{1}{0.6} = 1.67$$

Re-deriving the IS Curve

$$Y = C + I + G + EX - IM$$

Divide through by
potential GDP

$$\frac{Y}{\bar{Y}} = \frac{C}{\bar{Y}} + \frac{I}{\bar{Y}} + \frac{G}{\bar{Y}} + \frac{EX}{\bar{Y}} - \frac{IM}{\bar{Y}}$$

Substitute for all
these using our rules

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Then subtract one
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from both sides

$$\frac{Y - \bar{Y}}{\bar{Y}} \equiv \tilde{Y} = \bar{a}_c + mpc \cdot \tilde{Y} + \bar{a}_i - \bar{b}(R - \bar{r}) + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} - 1$$

$$\tilde{Y} = \bar{a}_c + mpc \cdot \tilde{Y} + \bar{a}_i - \bar{b} (R - \bar{r}) + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} - 1$$

Combine $\tilde{Y}$ terms on the left-hand side.

$$\tilde{Y} - mpc \cdot \tilde{Y} = \bar{a}_c + \bar{a}_i - \bar{b} (R - \bar{r}) + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} - 1$$

$$(1 - mpc) \cdot \tilde{Y} = \bar{a}_c + \bar{a}_i - \bar{b} (R - \bar{r}) + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} - 1$$

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The right-hand side is just the standard IS curve. Simplify

$$(1 - mpc) \cdot \tilde{Y} = \bar{a} - \bar{b} (R - \bar{r})$$

Now divide through by $(1 - mpc)$

$$\tilde{Y} = \frac{1}{1 - mpc} \times [\bar{a} - \bar{b} (R - \bar{r})]$$

What does the multiplier do?

$$\tilde{Y} = \frac{1}{1 - mpc} \times [\bar{a} - \bar{b} (R - \bar{r})]$$

- It enhances the power of government's fiscal and monetary policies.
(It also amplifies negative shocks)
- Think of construction workers who get jobs building new bridges funded by a big increase in government spending, G
- This is a **shift in the IS curve**, where $\bar{a}$ increases to $\bar{a}' = 0.01$, while $R = \bar{r}$
- There is a multiplier effect if the construction worker spends mpc of his or her current income rather than saving it

$$\tilde{Y}' = \frac{1}{1 - mpc} \times [\bar{a}' - \bar{b} (R - \bar{r})]$$

$$\tilde{Y}' = \frac{1}{1 - 0.4} \times 0.01 \approx 0.0167$$

- Rather than increase GDP by 1%, government spending increases it by 1.67%

IS curve

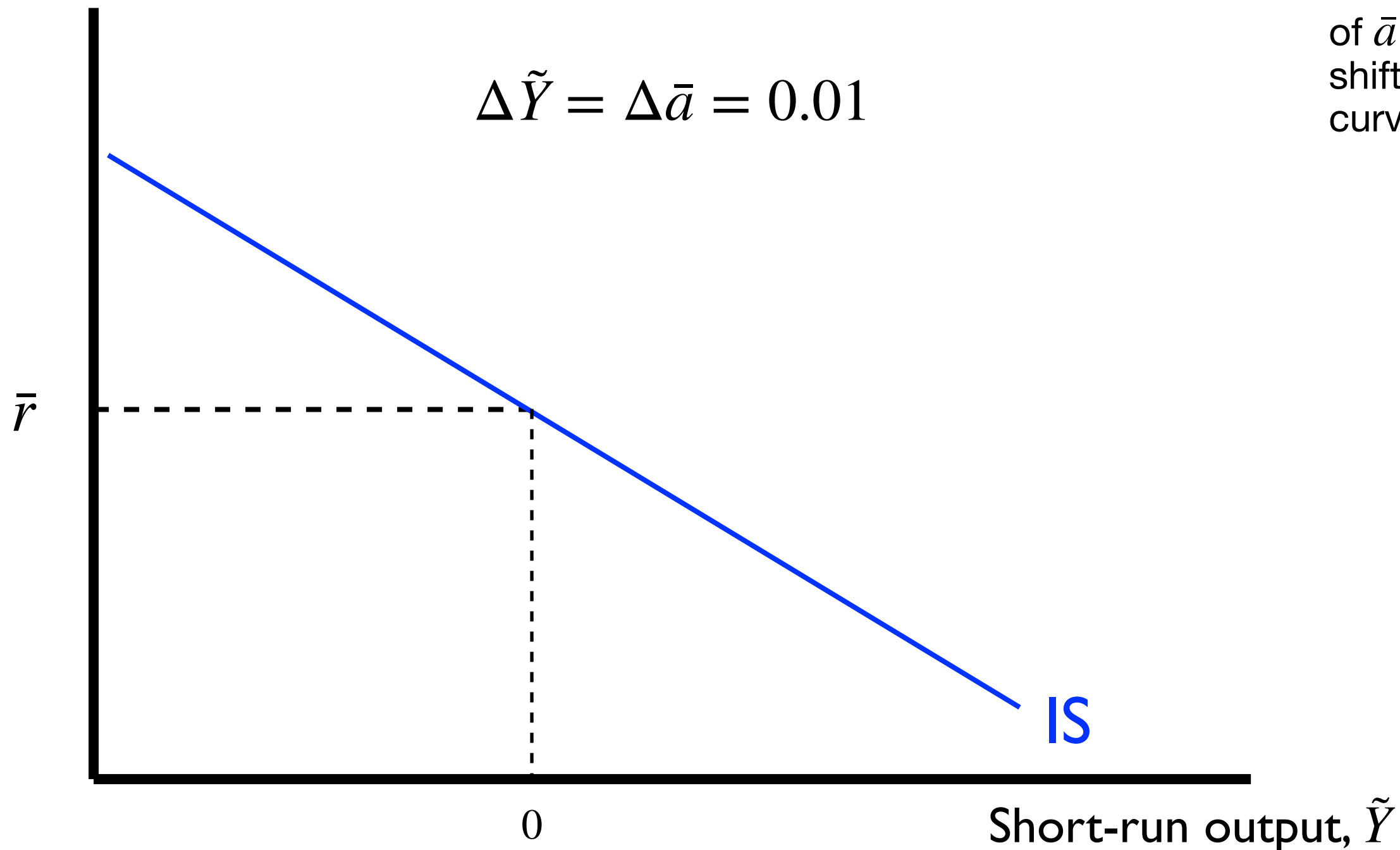
$$\tilde{Y} = \bar{a} - \bar{b}(R - \bar{r})$$

Suppose G rises by 1%

- A positive demand shock of $\bar{a}' = 0.01$ shifts the IS curve rightward

$$\Delta \tilde{Y} = \Delta \bar{a} = 0.01$$

Real interest rate, R



IS curve

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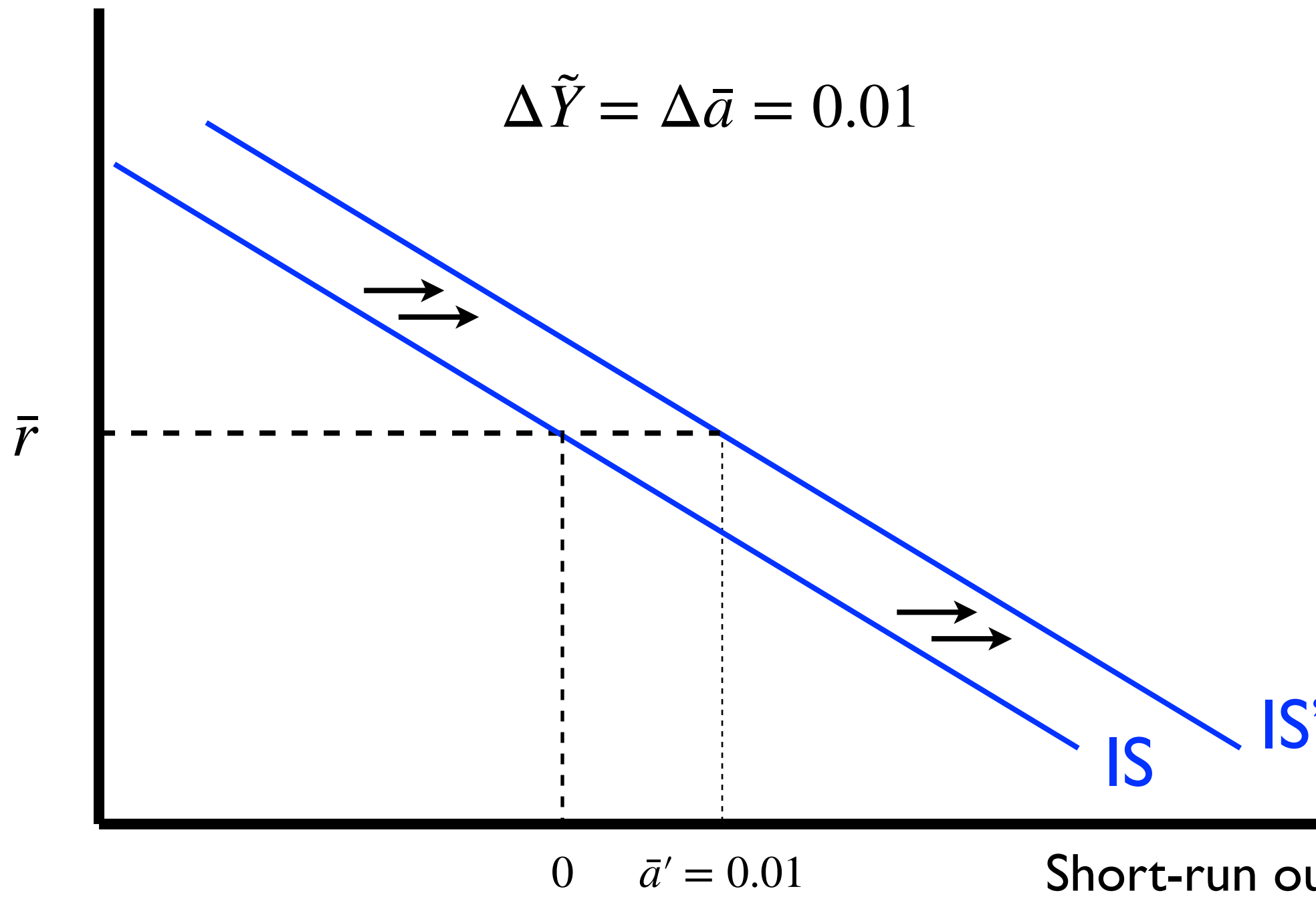
- If monetary policy does not react, $R = \bar{r}$

- And

$$\Delta \tilde{Y} = \Delta \bar{a} = 0.01$$

In words: a 1% increase in G produces a 1% increase in Y

Real interest rate, R



IS curve

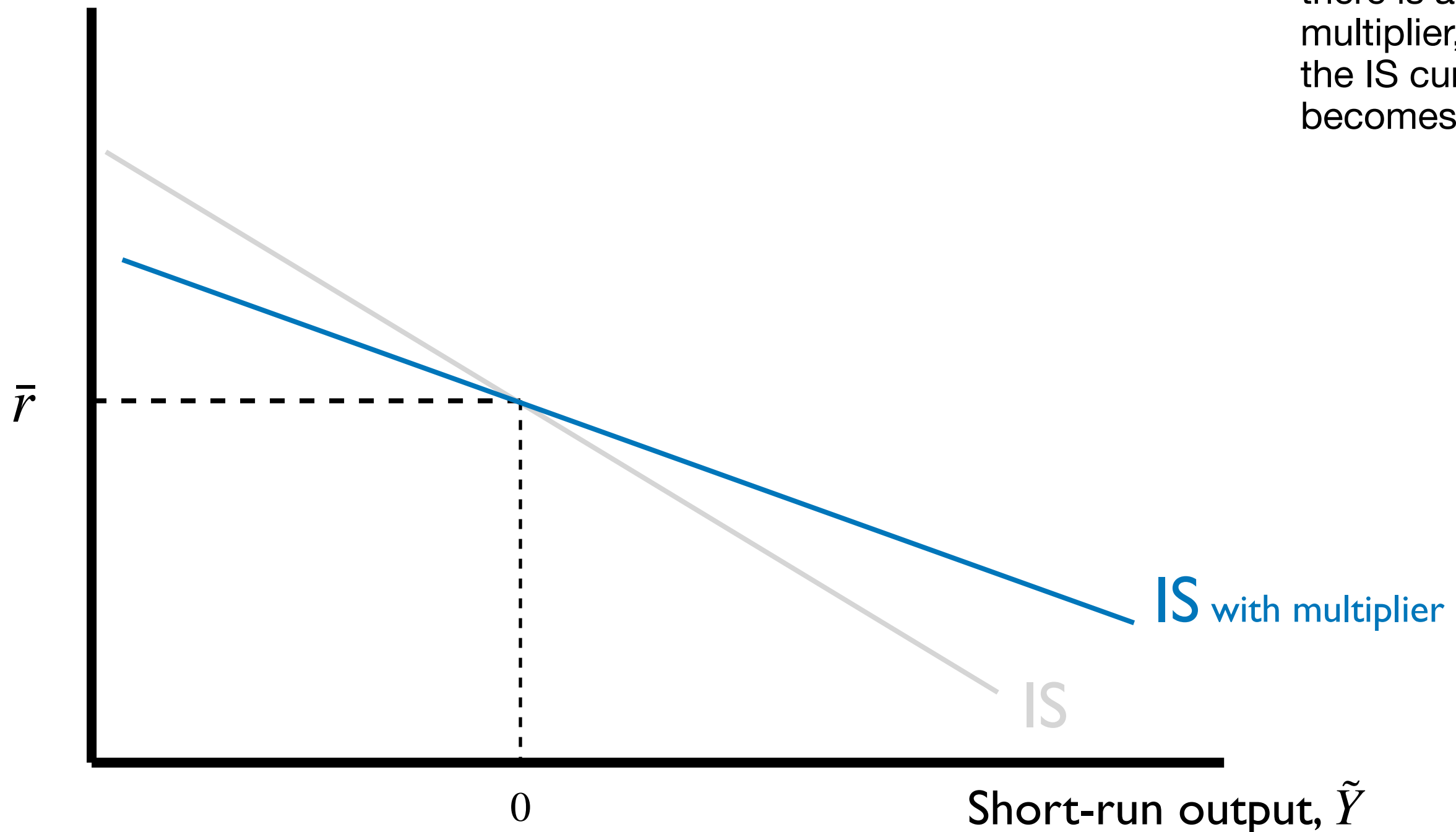
$$\tilde{Y} = \bar{a} - \bar{b} (R - \bar{r})$$

IS curve with multiplier

$$\tilde{Y} = \frac{1}{1 - mpc} \times [\bar{a} - \bar{b} (R - \bar{r})]$$

- When $mpc > 0$, there is a multiplier, and the IS curve becomes **flatter**

Real interest rate, R



IS curve

$$\tilde{Y} = \bar{a} - \bar{b}(R - \bar{r})$$

IS curve with multiplier

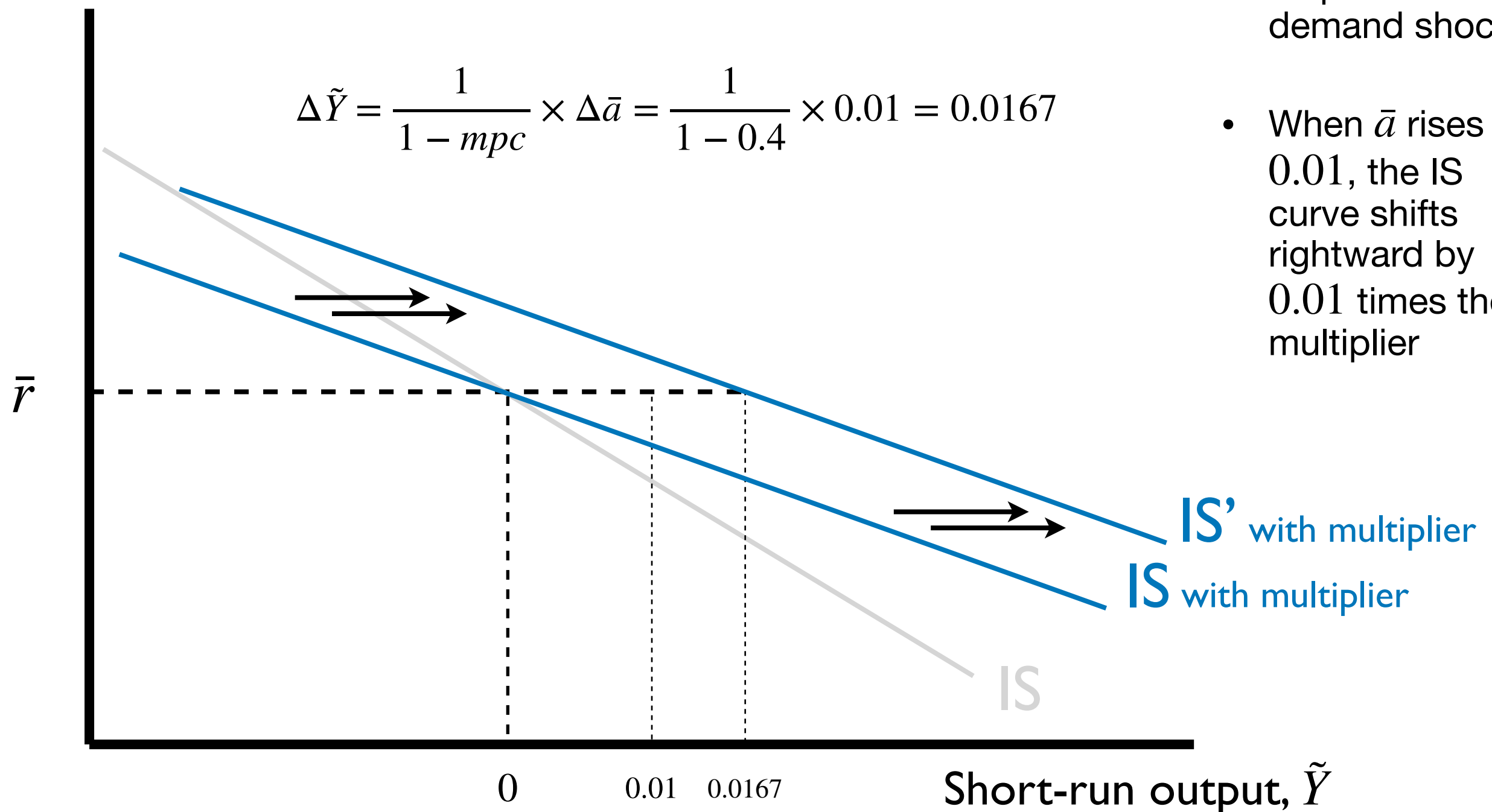
$$\tilde{Y} = \frac{1}{1 - mpc} \times [\bar{a} - \bar{b}(R - \bar{r})]$$

- A multiplier also amplifies demand shocks

- When $\bar{a}$ rises by 0.01, the IS curve shifts rightward by 0.01 times the multiplier

$$\Delta \tilde{Y} = \frac{1}{1 - mpc} \times \Delta \bar{a} = \frac{1}{1 - 0.4} \times 0.01 = 0.0167$$

Real interest rate, R



IS curve

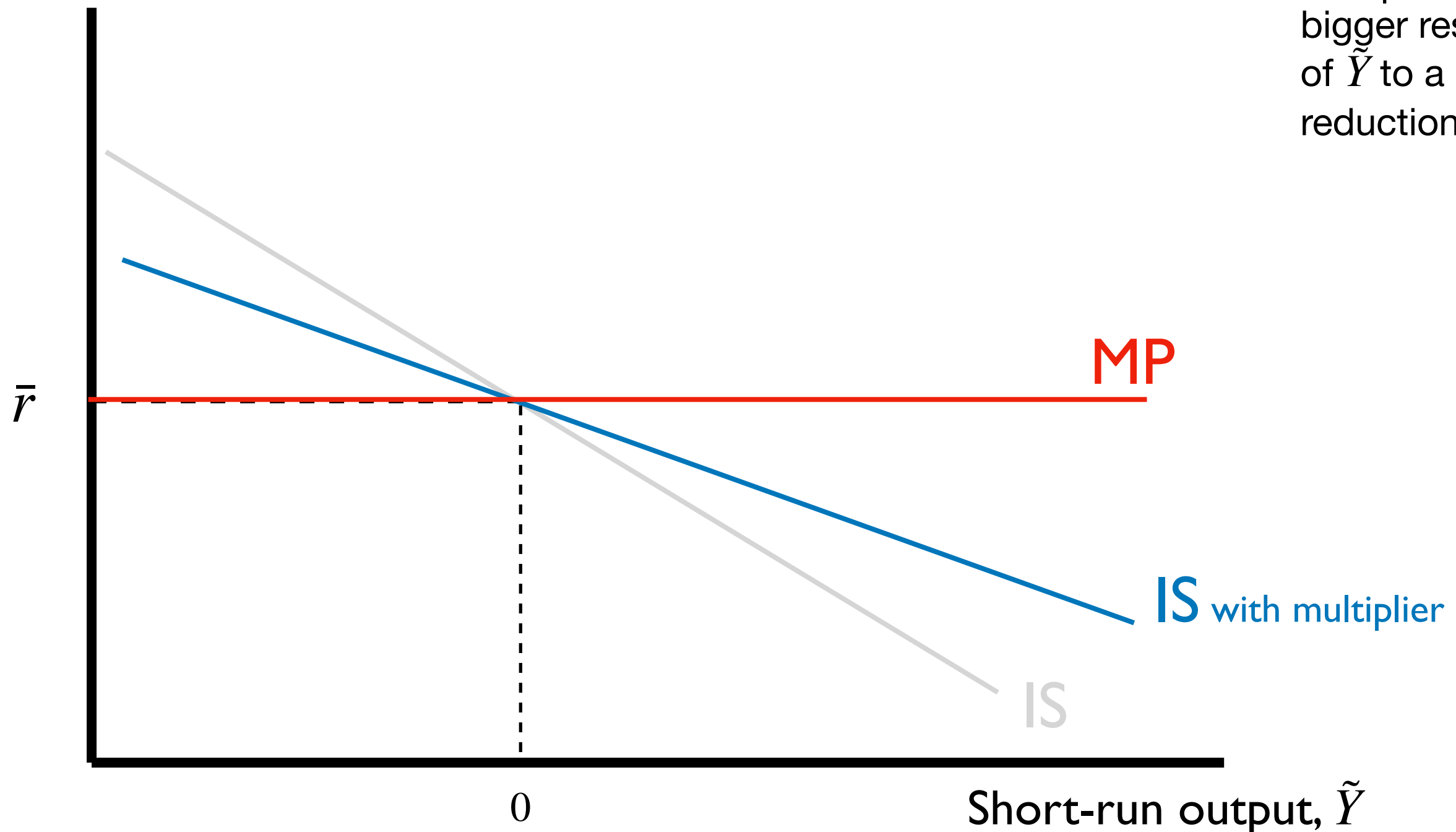
$$\tilde{Y} = \bar{a} - \bar{b} (R - \bar{r})$$

IS curve with multiplier

$$\tilde{Y} = \frac{1}{1 - mpc} \times [\bar{a} - \bar{b} (R - \bar{r})]$$

- A flatter IS curve also produces a bigger response of $\tilde{Y}$ to a reduction in R

Real interest rate, R



IS curve

$$\tilde{Y} = \bar{a} - \bar{b}(R - \bar{r})$$

IS curve with multiplier

$$\tilde{Y} = \frac{1}{1 - mpc} \times [\bar{a} - \bar{b}(R - \bar{r})]$$

- A flatter IS curve also produces a bigger response of $\tilde{Y}$ to a reduction in R

Real interest rate, R

